

BHTTP/1.0 — Binary HTTP

Course project · HTTP, in binary · two tracks, one protocol · now **you** write the spec | Java ref: bserve ./www 9000 · bcurl -v localhost:9000/index.html

1 Transport & framing

TCP, **one persistent connection**. All integers unsigned **big-endian** (network order). A frame = **9-byte header** + exactly **Length payload bytes**: readFully(9) then readFully(Length) — partial TCP reads are normal. **MAX_FRAME = 16,777,215** ($2^{24}-1$); larger → framing error → close. Timeouts: server idle 60s, client connect 5s / read 10s.

```
0 1 2 3
0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1
+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+
| Payload Length (24) |
+-----+-----+-----+-----+-----+-----+-----+-----+
| Type (8) | Flags (8) |R| Request-ID (31)|
+-----+-----+-----+-----+-----+-----+-----+-----+
| Payload (Length bytes) ...
```

Field	Width	Rule / defense (why 24 / 8 / 8 / 31?)
Length	24	Payload-only length. 16-bit fragments every image; 32-bit invites 4 GiB DoS alloc. 24 fits a small file / large chunk, bounds memory, and drives unknown-skip .
Type	8	0x01 REQUEST, 0x02 RESPONSE, 0x03 DATA — 3 used, 253 spare for v2 . 4 bits leaves no room; 16 wastes a byte/frame.
Flags	8	Per-type. END_STREAM=0x01, END_HEADERS=0x02. Reserved bits MUST be 0 on send, ignored on receive.
R + ID	1+31	Top bit MUST be 0 (keeps Java int ≥ 0). ID 0 illegal in v1. Client odd 1, 3, 5..., server echoes. Catches mismatch + allows future mux w/o header change.

A receiver meeting a frame type it does not know MUST skip it cleanly: read + discard exactly Length bytes, ignore flags/ID, continue — MUST NOT close. That is how we leave room for a version 2.

2 Frame payloads

REQUEST 0x01: method:u8 + pathLen:u16 + path:UTF-8 + numHeaders:u16 + headerBlock. method: 0x01 GET, 0x02 HEAD, else **400**. path **MUST** start /, no NUL, ≤2048 B. Flags **MUST** be 0x03 (END_STREAM|END_HEADERS) in v1 (no request body). **RESPONSE** 0x02: status:u16 + numHeaders:u16 + headerBlock, status 100–599. END_STREAM=1 → no body; 0 → ≥1 DATA follows. **DATA** 0x03: opaque file bytes (Length bytes); END_STREAM=1 on last chunk else 0. Empty file / HEAD → RESPONSE END_STREAM=1, zero DATA frames.

3 Headers — HPACK mechanisms 1 + 2 in an evening

Static table (the ten names we actually send): 0x01 :path 0x02 :status(res.) 0x03 content-type 0x04 content-length 0x05 date 0x06 server 0x07 host 0x08 user-agent 0x09 accept 0x0A connection. Per entry, first byte = discriminator: 0x01–0x0A → index:u8 + valueLen:u16 + value; 0x00 → 0x00 + nameLen:u16 + name + valueLen:u16 + value (name lowercase [a-z0-9-]; receiver **MUST** accept unknown literals). Any other discriminator → **400**. Limits: ≤64 headers, name ≤256 B, value ≤8 KiB, block ≤64 KiB, else 400. No Huffman / dynamic table / variants in v1 — fixed u16 keeps scope while showing indexing + literal fallback.

4 Path → file & status

Strip #frag then ?query → percent-decode %XX (bad → **400**) → / → /index.html, trailing / appends index.html → normalize; **MUST** stay under root else **400** (attack, not 404). Symlink escaping root → 400. Regular file → **200** + content-type + content-length + date(IMF) + server:bserve/1.0, then DATA in 16 KiB chunks. Missing / dir-without-index → **404** with Not Found body (still exercises DATA). HEAD → headers (content-length=size), no DATA. MIME: html/htm text/html, css text/css, js text/javascript, json application/json, txt text/plain, png/jpeg/gif/svg image/*, else application/octet-stream. I/O failure mid-response → **500**, then close.

5 Connection lifecycle & interop

Server bserve <root> <port>: accept loop; per-conn loop read → dispatch → RESPONSE[+DATA] → flush → repeat. Semantic errors (bad method/path/headers/traversal) → **4xx RESPONSE and keep open**. Fatal framing (truncated header/payload, Length>MAX, ID==0, reserved-bit set) → try 400 if boundary known else TCP close — never rescan a binary stream. **Client** bcurl [-v] host:port/path: **exactly ONE Socket**; REQUEST id=1 (host, user-agent:bcurl/1.0, accept:*/*, connection:keep-alive); read until same-ID END_STREAM=1. DATA → stdout raw; -v dumps every frame → stderr. Exit **0** on 2xx/3xx, **1** on 4xx/5xx, **2** transport/usage. DATA-before-RESPONSE or ID mismatch → transport error. **Pairs**: only this spec crosses between implementations — a client that works solely against its own server is an implementation, not a protocol.

Wire example (GET /hello.txt → 200, 18 B). REQ hdr 00 00 40 01 03 00 00 00 01 (Len 64, REQUEST, flags 0x03, ID 1) + 01(GET) 00 0A /hello.txt 00 04 + 4 indexed hdrs. RESP hdr 00 00 51 02 00 00 00 00 01 (Len 81, RESPONSE, more DATA, ID 1) + 00 C8(200) 00 04 + 4 hdrs. DATA hdr 00 00 12 03 01 00 00 00 01 (Len 18, END_STREAM) + hello from bserve\n. Full byte-by-byte annotation: see HEXDUMP.md.